Anytime-Valid Nonparametric Testing of Synergistic (Group) Causality, with an Application Establishing Its Absence in Daily Financial Markets

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Abstract

Does a *group* of causes act on an effect jointly—beyond the sum of its members' separate contributions—and can we decide this *as data arrive*, stopping the moment the evidence is conclusive? Every existing test of such synergy is fixed-sample: sufficient-cause interaction indices, joint-effect estimators, and information-theoretic decompositions (PID, SURD, PEID) all return a point value at a pre-committed sample size, with no valid way to monitor evidence continuously. We introduce ANTE, an *anytime-valid* nonparametric test of a super-additive synergy null built on testing-by-betting. A bounded, mean-centred interaction score is accumulated into a nonnegative capital process (an e-process); Ville's inequality yields time-uniform Type-I error control under optional stopping and continuous monitoring, requiring neither stationarity nor correct model specification. We prove validity, a growth (consistency) theorem giving power one and an explicit detection-time bound under the alternative, and extend the construction to $\boldsymbol{k}$-way groups and to an out-of-sample predictive (Granger) synergy for time series. Empirically, ANTE holds its level under continuous monitoring where a peeking fixed-sample test inflates tenfold, matches the best batch estimators in detection accuracy, and—uniquely—isolates genuine interaction from mere joint dependence. Deployed as a demonstrably powered instrument, ANTE *establishes* that super-additive group causality is *absent* from daily financial markets across thousands of asset triples, while it detects the turbulent energy cascade in physics. Synergistic group causality is real, but it is not a daily-markets phenomenon.

Keywords: anytime-valid inference, e-processes, testing by betting, synergy, group causality, Granger causality

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1 Introduction

Some effects require a coalition. Two drugs that are inert alone can be lethal together; two genes silent in isolation can jointly switch on a phenotype; two market shocks, each absorbed on its own, can compound into a crisis. The common structure is *synergy*: the joint action of a set of causes exceeds the sum of their individual actions. Detecting synergy—distinguishing "you need *both* A and B" from "A helps and B helps, separately"—is a recurring scientific question, and a surprisingly subtle one, because it is a statement about an *interaction*, not about any single cause.

The question has been formalised at least three times, in mutually unaware literatures. In epidemiology, Rothman's sufficient-component-cause model (Rothman 1976) defines synergy as two factors being component causes of the same causal "pie", measured by the additive-scale interaction contrast and its relative-excess-risk index (Hosmer and Lemeshow 1992; VanderWeele and Knol 2014), with a complete counterfactual axiomatisation for arbitrarily many binary causes due to VanderWeele and Richardson (2012) and, very recently, a multi-factor generalized synergy index (La Torre and D'Urso 2026). In statistics, a parallel programme estimates the effect of a *joint* intervention $\mathrm{do}(\{A, B, \dots\})$ by fusing experiments (Roquero Gimenez and Rothenhäusler 2023) or identifying the joint law of potential outcomes (Wu and Mao 2025). In information theory, Partial Information Decomposition (Williams and Beer 2010) and its dynamical descendants SURD (Martínez-Sánchez et al. 2024) and PEID (Yang et al. 2026) split the information that sources carry about a target into redundant, unique, and *synergistic* parts.

All three share one limitation that this paper is written to remove: they are **fixed-sample**. Each returns a point estimate—and, at best, a confidence interval—at a sample size chosen *before* looking at the data. None supports the mode of inference that modern data actually demand: watch the evidence accumulate, and stop as soon as it is decisive, without paying a multiplicity penalty for having looked. A practitioner monitoring a streaming system who applies a fixed-sample synergy test repeatedly, "peeking" after each new observation, silently destroys the guarantee the test was built to provide; we show below that the Type-I error of exactly this natural procedure inflates from a nominal 5% to over 50%.

The machinery to do continuous monitoring correctly now exists. Testing by betting (Shafer 2021), e-processes and safe anytime-valid inference (Ramdas et al. 2023; Grünwald et al. 2024), and confidence sequences (Howard et al. 2021) provide tests whose validity holds *simultaneously at all sample sizes* and under *any* data-dependent stopping rule. The idea is disarmingly simple: a sceptic bets against the null; if the null is true the sceptic's capital is a nonnegative martingale and cannot grow, so large capital is itself evidence against the null, quantified by Ville's inequality (Ville 1939). This programme has produced anytime-valid tests for means (Waudby-Smith and Ramdas 2024), off-policy and average treatment effects (Waudby-Smith et al. 2024), and other structured nulls—but, to our knowledge, *never for an interaction or synergy contrast.* The two bodies of work—synergy on one side, anytime-valid inference on the other—have not met.

Contributions.

We make them meet. This paper contributes:

1. **A test.** ANTE (*Anytime-valid Nonparametric Test of synErgy*): a betting-martingale test of a super-additive synergy null. Under a randomized or known design we build a bounded interaction score that is exactly mean-centred under H_0 (Lemma 1); accumulating it as capital yields a nonnegative martingale (Theorem 2) and, via Ville's inequality, time-uniform Type-I control under arbitrary optional stopping (Corollary 3). Validity needs no asymptotics, no variance assumption, and no stationarity.
2. **Theory of power.** We prove that under any fixed super-additive alternative the capital grows exponentially, so the test rejects with probability one (Theorem 4), and we give an explicit high-probability bound on the detection time (Proposition 5). Validity without power is vacuous—a test that never rejects controls Type-I error trivially—so the two guarantees must be proved together; the detection-time bound further quantifies the price of monitoring, scaling only logarithmically in the level α.
3. **Groups and time series.** We extend the score to k-way groups by contrasting a joint predictor carrying all higher-order interaction terms against a purely additive one (Proposition 6), and we instantiate a sequential test of *synergistic Granger causality*—super-additive out-of-sample predictive gain—for nonstationary time series, using forgetting recursive-least-squares predictors and a game-theoretic (prequential) null that dispenses with correct specification.
4. **A discrimination guarantee.** We show, and verify, that ANTE separates genuine irreducible interaction from two impostors that fool existing estimators: additive-but-both-needed dependence, and within-variable nonlinearity $Y = g(A)$ (Proposition 7).
5. **A scientific finding.** Turning the test on real data, we *establish*—as evidence of absence, not absence of evidence—that super-additive group causality is absent from daily financial markets. Across more than three thousand asset triples spanning returns, volatility and tail-stress, daily and hourly, equities, sector ETFs, macro assets and crypto, and 2-, 3- and 4-way groups, the financial evidence sits on a calibrated true-null distribution (median $\log_{10} E = -0.93$: a bet on synergy systematically loses), with zero family-wise detections. The very same test detects synthetic synergy at 74–100% power and recovers the turbulent energy cascade from fluid dynamics. Daily markets are common-factor plus pairwise; their irreducible higher-order structure is negligible—a quantified caution to in-sample synergy-in-finance claims (Scagliarini et al. 2020).

The paper is organised as follows. Section 2 surveys, in depth, the four literatures ANTE draws together—sufficient-cause interaction, joint-effect estimation, information-theoretic decomposition, and game-theoretic sequential inference—and states precisely the gap it fills. Section 3 fixes the model and defines the synergy null on two scales. Section 4 is the theoretical core: the interaction score, the betting e-process, validity, power, the k-way extension, the isolation guarantee, and the algorithms that realise them. Section 5 reports synthetic validation, a head-to-head benchmark against 2024–2026 methods, and the real-data study culminating in the evidence-of-absence finding. Section 6 discusses implications, limitations, and future work, and concludes; proofs are collected in the appendix.

2 Related work and positioning

Group causality—the notion that a set of causes acts on an effect jointly, beyond the sum of its members—has been studied, largely independently, in four research traditions. Three formalise *what synergy is* (epidemiological sufficient-cause interaction, statistical joint-effect estimation, and information-theoretic decomposition); the fourth supplies the inferential machinery we import (game-theoretic sequential testing). We review each in turn, emphasising what each offers and the single feature none of the first three provides—a valid sequential test—before stating the gap precisely in §2.6.

2.1 Sufficient-cause and additive-scale interaction

The oldest and most developed formalisation of "A and B together cause C" comes from epidemiology. Rothman's *sufficient-component-cause* ("causal pie") model (Rothman 1976) defines synergy as two factors being component causes of a common sufficient cause, so that their joint presence produces an effect exceeding the sum of their separate contributions. For two binary causes with counterfactual cell means $\mu_{ab} = \mathbb{E}[C(a, b)]$, this is captured on the additive (risk-difference) scale by the interaction contrast $\theta = \mu_{11} - \mu_{10} - \mu_{01} + \mu_{00}$; the corresponding estimand and its sampling theory—the *relative excess risk due to interaction* (RERI)—were placed on a firm inferential footing by Hosmer and Lemeshow (1992) and systematised in the widely used tutorial of VanderWeele and Knol (2014). The definitive modern treatment is the counterfactual axiomatisation of VanderWeele and Richardson (2012), which gives empirical and counterfactual conditions for both sufficient-cause interaction and the stronger *singular* interaction among an arbitrary number of dichotomous exposures; this is the batch null our sequential test most directly generalises.

On the estimation side, mechanistic (as opposed to merely statistical) interaction has its own measures: the peril-ratio index of synergy PRISM of Lee (2013), and its extension to gene–gene interaction under independence and Hardy–Weinberg assumptions (Lin and Lee 2018), which supply case-control tests for sufficient-cause interaction in the non-rare-disease regime. Most recently, La Torre and D'Urso (2026) generalise Rothman's two-factor synergy index S to an arbitrary number of dichotomous factors, reflecting active current interest in *multi-way* synergy. The common thread—and the common limitation—is that every method in this tradition is **fixed-sample**: a point estimate and confidence interval computed at a pre-specified n. The primary sources themselves observe that multi-way interaction estimates become unstable at small n; this is precisely the regime in which a sequential procedure that spends evidence adaptively, and stops as soon as it is decisive, would help most.

2.2 Joint effects, causal aggregation, and multi-cause identification

A second, more recent, statistical tradition asks not *whether* causes interact but *what the effect of intervening on several at once is.* Roquero Gimenez and Rothenhäusler (2023) develop *causal aggregation*: estimation and inference for the effect of a joint intervention $\mathrm{do}(\{A, B, \dots\})$ by constraint-based fusion of experiments that each manipulate only a few variables—directly a "sum-of-parts versus joint" question, but posed as effect estimation rather than a synergy test. Wu and Mao (2025) pursue nonparametric identification of the *joint distribution* of potential outcomes from multiple experiments, the closest existing analogue of a "probability of joint causation"; and deep structural-equation approaches such as the causal-graph normalising-flow models of Balgi et al. (2024) estimate interventional and counterfactual distributions from which joint effects could in principle be read off. These methods are powerful and general, but they share the fixed-design, one-shot character of §2.1: none provides a sequentially valid procedure with time-uniform coverage for a joint effect, even though—as §2.5 notes—the corresponding *single*-treatment anytime-valid theory is by now well developed (Waudby-Smith et al. 2024).

2.3 Information-theoretic synergy and higher-order interactions

The third tradition measures synergy as an *information* quantity. Interaction information (equivalently co-information; McGill 1954) was the first attempt to isolate the information three or more variables share beyond their pairwise relations, but it can be negative and conflates synergy with redundancy. *Partial Information Decomposition* (PID; Williams and Beer 2010) resolved this by decomposing the total information two sources carry about a target into non-negative redundant, unique, and *synergistic* atoms—the canonical modern definition of information-theoretic synergy. Because the PID lattice grows combinatorially and, as Lyu et al. (2026) show, becomes internally inconsistent for four or more variables, subsequent work has sought tractable, well-behaved surrogates. The O-information of Rosas et al. (2019) gives a single scalar that sign-indicates whether a system is net-synergistic or net-redundant and scales to many variables; it is a natural higher-order baseline. For *dynamical, causal* systems, two 2024–2026 methods are the direct state of the art. SURD (Martínez-Sánchez et al. 2024) decomposes transfer-entropy-style causality into synergistic, unique, and redundant components, robust to nonlinearity, colliders, and exogenous influence, and validated on canonical fluid-dynamics benchmarks. PEID (Yang et al. 2026) provides an *interventional* analogue of PID: under an independent maximum-entropy intervention that renders the sources independent, source-side redundancy vanishes and synergy is read off directly as $EI(A, B \to C) - \sum_i EI(T^{(i)} \to C)$; its *hierarchical additivity* property—synergy accumulates additively across partition refinements—is exactly the increment structure our subset-refinement e-process exploits (§4.5). These are the strongest existing synergy quantifiers and we benchmark against them directly (§5.2). Yet all of them are *estimators*, not tests: each returns a point synergy value with no null distribution, p-value, calibrated error rate, or stopping rule, so there is no principled way to declare "the evidence for synergy is now sufficient"—still less to do so under continuous monitoring.

2.4 Synergy and causality in time series and finance

Because our flagship application is financial, we separately review the time-series thread. Directed predictability is classically formalised by Granger causality (Granger 1969) and its multivariate/conditional extension (Geweke 1982), and, in the information-theoretic idiom, by transfer entropy (Schreiber 2000); the two coincide for Gaussian variables (Barnett et al. 2009), which is what lets us read our predictive synergy contrast as an interaction of conditional transfer entropies. Crucially, Stramaglia et al. (2014) show that pairwise Granger analysis *misses* synergy and that conditioning is required to reveal it—direct motivation for a genuinely joint test. In finance specifically, Scagliarini et al. (2020) compute PID synergy over triplets of global market indices and report synergistic information transfer, while the econometric-standard connectedness framework of Diebold and Yılmaz (2014) measures system-wide spillovers through variance decompositions. These works establish that synergy has been *measured* in markets, but always in-sample, at fixed sample size, and without an out-of-sample or error-controlled test—exactly the claims our evidence-of-absence analysis (§5.4) revisits under a stricter criterion.

2.5 Testing by betting and anytime-valid inference

The inferential machinery we import is game-theoretic statistics (Ramdas et al. 2023). Its foundation is Ville's inequality for nonnegative (super)martingales (Ville 1939); its modern reframing is testing by betting (Shafer 2021) and safe testing (Grünwald et al. 2024), in which a sceptic wagers against the null and large accumulated capital is evidence against it. The practical construction we build on is the hedged-capital / GRAPA betting strategy for bounded means of Waudby-Smith and Ramdas (2024), complemented by time-uniform confidence sequences (Howard et al. 2021) and the universal-inference likelihood ratio of Wasserman et al. (2020). Continuous-monitoring validity is what distinguishes these from classical tests, whose guarantees collapse under repeated looking—a failure quantified in the industrial A/B-testing setting by Johari et al. (2022). When many hypotheses are monitored at once, e-values (unlike p-values) can be merged by averaging under arbitrary dependence and by multiplication under independence (Vovk and Wang 2021), and support false-discovery-rate control via e-BH (Wang and Ramdas 2022)—the tools we use for family-wise/FDR control in the group scan. The closest causal application of this machinery is anytime-valid off-policy and average-treatment-effect inference (Waudby-Smith et al. 2024); like all prior work in this line it targets a *single* effect, never an interaction or synergy contrast.

2.6 The gap, and our position

The pattern in Table 1 is stark: the literatures that formalise synergy (§§2.1–2.4) are uniformly fixed-sample, while the sequential/anytime-valid literature (§2.5) has been developed only for single effects. A focused prior-art search (terms: *anytime-valid interaction test*, *e-process causal interaction*, *sequential test synergy*, *confidence sequence*

Table 1 The synergy literatures and the anytime-valid literature do not intersect. Every prior method either quantifies synergy but is fixed-sample, or is anytime-valid but targets only single effects. ANTE occupies the empty cell.

Literature	Handles synergy?	Anytime-valid / sequential?
Sufficient-cause / RERI / synergy index (§2.1)	yes	no (fixed n)
Joint effects / causal aggregation / joint PO (§2.2)	yes (joint effect)	no (one-shot)
PID / SURD / PEID / O-information (§2.3)	yes (info scale)	no (point estimate)
Time-series / finance synergy (§2.4)	yes (in-sample)	no (batch)
Betting / e-processes / anytime-valid ATE (§2.5)	no (single effects)	yes
ANTE (this paper)	**yes**	**yes**

interaction effect, *test martingale synergistic*) returned no sequential, anytime-valid test whose null is "the joint causal effect of a set of causes equals the sum of its parts." Related betting tests exist for other structured nulls—stochastic dominance, independence, off-policy means—but not for an additive or multiplicative interaction contrast. ANTE is, to our knowledge, the first test to place synergy detection on an anytime-valid footing, and it does so on both the additive causal-pie scale and the information-theoretic scale, reducing them to a single testable null.

3 Problem setup and the synergy null

We develop the null on two scales—an interventional additive-interaction scale and an observational predictive scale—and show they reduce to the same qualitative statement: *joint equals sum of parts.*

3.1 Interventional synergy

Observations arrive as a stream $Z_1, Z_2, \ldots$ with $Z_t = (X_t, \mathbf{T}_t, C_t)$, where $C_t \in [0,1]$ is a bounded effect at a target, $\mathbf{T}_t = (T_t^{(1)}, \ldots, T_t^{(k)}) \in \{0,1\}^k$ are candidate binary causes, and X_t are optional covariates. Let $\mathcal{F}_t = \sigma(Z_1, \ldots, Z_t)$ be the natural filtration. We adopt the potential-outcomes model with $C_t(\mathbf{t})$ the outcome under $\mathrm{do}(\mathbf{T} = \mathbf{t})$ and cell means $\mu_{\mathbf{t}} = \mathbb{E}[C(\mathbf{t})]$.

Assumption 1 (Design) Per step, (i) *consistency* $C_t = C_t(\mathbf{T}_t)$; (ii) *positivity* $\pi(\mathbf{t} \mid X) \geq \pi_{\min} > 0$; (iii) *unconfoundedness* $\mathbf{T}_t \perp \{C_t(\mathbf{t})\} \mid X_t$. Under a randomized or known design $\pi(\mathbf{t} \mid X) = \pi(\mathbf{t})$ is known and (i)–(iii) hold by construction.

Definition 1 (Additive interaction contrast) For two causes A, B,

$$\theta \;=\; \mu_{11} - \mu_{10} - \mu_{01} + \mu_{00}.$$

$\theta = 0$ states that the joint effect of (A, B) equals the sum of the two individual effects (*no synergy*); $\theta > 0$ is super-additive synergy (a causal-pie / AND mechanism), $\theta < 0$ is antagonism. The one-sided synergy null is $H_0 : \theta \leq 0$ (two-sided $H_0 : \theta = 0$).

The information-theoretic synergy of §2.3 targets the same idea in bits: under an independent maximum-entropy intervention, source-side redundancy vanishes and $\mathrm{Syn}_{EI} = EI(A, B \to C) - \sum_i EI(T^{(i)} \to C)$ (Yang et al. 2026). Both scales reduce to *joint = sum of parts*; §4 develops the exactly-identified, bounded additive-scale test in full and remarks on the information-scale instantiation.

3.2 Observational (Granger) synergy for time series

For financial and other observational streams there is no intervention, so we move to an out-of-sample predictive definition in the spirit of Granger (Granger 1969). Let the (standardized) panel have target Y, candidate sources A, B, optional conditioning set Z (e.g. a market factor), and lag order p. For $S \subseteq \{A, B\}$ let $L(S)$ be the expected one-step-ahead loss of the best predictor of Y_t from the $\{Y, Z\}$-past augmented by the lagged histories of S, and let $\Delta(S) = L(\varnothing) - L(S)$ be its predictive gain.

Definition 2 (Predictive (Granger) synergy)

$$\mathrm{Syn}(A, B \to Y) \;=\; \Delta(\{A, B\}) - \Delta(\{A\}) - \Delta(\{B\}).$$

$\mathrm{Syn} > 0$ means the joint predictor beats the sum of the marginal predictors: the pair carries predictive structure neither member carries alone.

For Gaussian variables, Granger causality equals transfer entropy (Barnett et al. 2009), so Syn is an interaction of conditional transfer entropies—the same object SURD/PID call synergistic information, recast as an out-of-sample predictive contrast. The advantage of the predictive framing is that it supports a *prequential* null (Dawid 1984) requiring neither stationarity nor correct specification, as we now make precise.

4 The ANTE test

4.1 A per-sample score, centred under the null

Under Assumption 1 define the inverse-probability-weighted *interaction score*

$$\psi_t \;=\; c(A_t, B_t)\,\frac{C_t}{\pi(A_t, B_t)}, \qquad c(a, b) = (-1)^{(1-a)+(1-b)}, \tag{1}$$

so that $c(a, b) = +1$ on the concordant cells $(a, b) \in \{(1, 1), (0, 0)\}$ and $c(a, b) = -1$ on the discordant cells $(a, b) \in \{(1, 0), (0, 1)\}$. Thus ψ_t is the empirical 2×2 interaction contrast reweighted by the design.

Lemma 1 (Unbiasedness and boundedness) *Under Assumption 1, $\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] = \theta$. Hence under $H_0 : \theta = 0$, $\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] = 0$. Moreover $C_t \in [0, 1]$ and $\pi(a, b) \geq \pi_{\min}$ imply $|\psi_t| \leq c_\psi := 1/\pi_{\min}$; for a balanced 2×2 design $\pi_{\min} = \frac{1}{4}$ and $\psi_t \in [-4, 4]$.*

Proof By consistency and unconfoundedness, $\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] = \sum_{a,b} \pi(a,b)\, c(a,b)\, \mu_{ab}/\pi(a,b) = \sum_{a,b} c(a,b)\mu_{ab} = \mu_{11} - \mu_{10} - \mu_{01} + \mu_{00} = \theta$. The bound is immediate from $0 \le C_t \le 1$ and $\pi(a,b) \ge \pi_{\min}$. □

Boundedness—not any variance or tail assumption—is what makes the betting martingale below exactly valid. A doubly-robust (AIPW) variant that replaces ψ_t by $\phi_t = \sum_{a,b} c(a,b)\big[\frac{\mathbb{1}\{A_t=a,B_t=b\}}{\pi(a,b|X_t)}(C_t - \hat{\mu}_{ab}(X_t)) + \hat{\mu}_{ab}(X_t)\big]$ keeps $\mathbb{E}[\phi_t \mid X_t] = \theta(X_t)$ Neyman-orthogonal, so slowly-learned nuisances do not break centring; this is the route to the observational extension and is compatible with everything below because ϕ_t is again bounded and predictably centred.

4.2 The betting e-process

Fix a level $\alpha \in (0,1)$. Start with unit capital $E_0 = 1$ and bet a *predictable* fraction λ_t—a function of $\psi_1, \dots, \psi_{t-1}$ (equivalently $\mathcal{F}_{t-1}$-measurable) only:

$$E_t \;=\; \prod_{s=1}^{t}\big(1 + \lambda_s \psi_s\big), \qquad |\lambda_s| \le \frac{\gamma}{c_\psi} \;\; (\gamma < 1) \text{ so that } 1 + \lambda_s\psi_s > 0. \tag{2}$$

Theorem 2 (Validity) *Under any $P \in H_0$ (i.e. $\theta = 0$), the process $(E_t)_{t\ge 0}$ in (2) is a nonnegative martingale with $\mathbb{E}_P[E_t] = 1$ for all t.*

Proof Non-negativity: each factor $1+\lambda_s\psi_s > 0$ by the bound on λ_s and Lemma 1. Martingale property: λ_t is $\mathcal{F}_{t-1}$-measurable and, by Lemma 1 under H_0, $\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] = 0$, so

$$\mathbb{E}[E_t \mid \mathcal{F}_{t-1}] = E_{t-1}\big(1 + \lambda_t\, \mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}]\big) = E_{t-1}.$$

Taking expectations gives $\mathbb{E}_P[E_t] = E_0 = 1$. □

Corollary 3 (Anytime-valid Type-I control) *For the rule "reject the first time $E_t \ge 1/\alpha$",*

$$\mathbb{P}_{H_0}\big(\exists\, t \ge 1 :\; E_t \ge 1/\alpha\big) \;\le\; \alpha.$$

Consequently the test controls Type-I error uniformly over all stopping times*: one may monitor after every observation, stop early, or peek indefinitely, and the guarantee holds. E_t is an e-value at every t, and $1/\max_{s\le t} E_s \wedge 1$ is an anytime-valid p-value.*

Proof (E_t) is a nonnegative martingale with $\mathbb{E}[E_0] = 1$ (Theorem 2), hence a nonnegative supermartingale; Ville's inequality (Ville 1939) gives $\mathbb{P}(\sup_t E_t \ge 1/\alpha) \le \mathbb{E}[E_0]\alpha = \alpha$. □

Remark 1 (Why peeking a fixed-sample test fails) A classical fixed-n test controls $\mathbb{P}(\text{reject at } n)$ for a single pre-committed n. Monitoring it continuously reports $\mathbb{P}(\exists t : \text{reject at } t)$, which is unbounded in the horizon: the more you look, the more likely a chance excursion crosses the fixed critical value. Corollary 3 is exactly the statement a peeking test lacks. We quantify the resulting inflation (5% → 50%) in §5.1.

4.3 Two-sided tests and the choice of bet

Synergy may be positive or negative (XOR gives $\theta = -2$). Run two capital processes—E_t^+ betting $\lambda_s \geq 0$ and E_t^- betting $\lambda_s \leq 0$—and average, $E_t = \frac{1}{2}(E_t^+ + E_t^-)$. An average of e-processes is an e-process (its expectation under H_0 is 1 by linearity), so the two-sided test is valid against $\theta \neq 0$.

Any predictable λ_t is valid; a *good* one grows capital fast under H_1, minimising expected detection time—the growth-rate-optimal (GRO) criterion (Grünwald et al. 2024). We use a truncated plug-in Kelly fraction from the running moments of ψ,

$$\lambda_t = \mathrm{clip}\Big(\frac{\hat{\mu}_{t-1}}{\hat{v}_{t-1}},\ \pm\frac{\gamma}{c_\psi}\Big), \quad \hat{\mu}_{t-1} = \tfrac{1}{t-1}\sum_{s<t}\psi_s, \quad \hat{v}_{t-1} = \tfrac{1}{t-1}\sum_{s<t}\psi_s^2, \tag{3}$$

the aGRAPA/GRAPA strategy of Waudby-Smith and Ramdas (2024) specialised to the interaction score, with a short warm-up ($\lambda_t = 0$ for $t \leq t_0$).

4.4 Power and detection latency

Validity is only half the story; a Q1 test must also be shown to *reject when it should.* We give a growth (consistency) theorem and an explicit latency bound. Consider the one-sided test with a fixed oracle bet $\lambda_t \equiv \lambda \in (0, \gamma/c_\psi]$ under an alternative with a persistent super-additive signal.

Assumption 2 (Persistent alternative) There is $\delta > 0$ with $\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] \geq \delta$ for all t, and $|\psi_t| \leq c_\psi$.

Theorem 4 (Consistency / power one) *Under Assumption 2, for any fixed $\lambda \in (0, \gamma/c_\psi]$ the growth rate is bounded below,*

$$\frac{1}{t}\ \log E_t \ \xrightarrow[t\to\infty]{a.s.}\ g(\lambda)\ \geq\ \lambda\delta - \tfrac{1}{2}\lambda^2 c_\psi^2\ >\ 0 \quad \textit{whenever}\ \ \lambda < \frac{2\delta}{c_\psi^2}.$$

Hence $E_t \to \infty$ almost surely and the anytime-valid test rejects with probability one; its power at level α tends to 1.

Sketch (full proof in Appendix A) Write $\log E_t = \sum_{s\leq t}\log(1+\lambda\psi_s)$. For $|\lambda\psi_s| \leq \gamma < 1$ the elementary bound $\log(1+x) \geq x - \frac{1}{2}x^2/(1-\gamma)$ and, more simply, $\log(1+x) \geq x - x^2$ on $[-\frac{1}{2}, \frac{1}{2}]$ give $\mathbb{E}[\log(1+\lambda\psi_s) \mid \mathcal{F}_{s-1}] \geq \lambda\delta - \frac{1}{2}\lambda^2 c_\psi^2$. The centred increments $\log(1+\lambda\psi_s) - \mathbb{E}[\cdot \mid \mathcal{F}_{s-1}]$ are bounded martingale differences, so a strong law for martingales gives $t^{-1}\log E_t \to g(\lambda)$ a.s. with $g(\lambda) \geq \lambda\delta - \frac{1}{2}\lambda^2 c_\psi^2$, strictly positive for $\lambda < 2\delta/c_\psi^2$. Thus $\log E_t \to +\infty$ and E_t eventually exceeds $1/\alpha$. □

Proposition 5 (Detection-time bound) *Under Assumption 2 with fixed $\lambda < 2\delta/c_\psi^2$ and growth rate $g = g(\lambda) > 0$, for any $\eta \in (0, g)$ there is a finite almost-surely stopping time after* *which* $\log E_t \geq (g-\eta)t$*; consequently the rejection time* $\tau_\alpha = \inf\{t : E_t \geq 1/\alpha\}$ *satisfies, with high probability,*

$$\tau_\alpha \;\leq\; \frac{\log(1/\alpha)}{g-\eta} \;\approx\; \frac{\log(1/\alpha)}{\lambda\delta - \frac{1}{2}\lambda^2 c_\psi^2}.$$

The bound is minimised near the Kelly bet $\lambda^\star = \delta/c_\psi^2$*, giving* $\tau_\alpha \lesssim 2c_\psi^2 \log(1/\alpha)/\delta^2$*: detection time scales like* δ^{-2} *in the synergy strength and only logarithmically in* $1/\alpha$*.*

The δ^{-2} scaling is the sequential analogue of the usual inverse-square-effect-size sample complexity, and it is borne out empirically: detection latency falls monotonically as synthetic synergy strengthens (§5.1).

4.5 k-way group synergy

For a group $\mathcal{G} = \{A_1, \ldots, A_k\}$ acting on Y, the natural null is that the joint predictor built from all of $\mathcal{G}$ offers no gain over a *purely additive* combination of the members' individual contributions. Let Φ^{add} be the feature map that stacks each member's own (linear and squared) lagged terms, and Φ^{joint} the map that additionally includes *all* higher-order interaction terms $\prod_{i\in\mathcal{S}} A_{i,t-\ell}$ over subsets $\mathcal{S} \subseteq \mathcal{G}$ with $|\mathcal{S}| \geq 2$. Let $\ell_t^{\text{add}}, \ell_t^{\text{joint}}$ be their out-of-sample losses and define the group score $s_t^{\mathcal{G}} = \ell_t^{\text{add}} - \ell_t^{\text{joint}}$.

Proposition 6 (Validity of the group test) *Let* $u_t = \operatorname{clip}(s_t^{\mathcal{G}}/\hat\sigma_{t-1}, -b, b)$ *with predictable scale* $\hat\sigma_{t-1}$ *and predictable* $\lambda_t \in [0, \gamma/b]$*. Under the group null* $H_0^{\mathcal{G}} : \mathbb{E}[s_t^{\mathcal{G}} \mid \mathcal{F}_{t-1}] \leq 0$ *(no super-additive group gain),* $E_t = \prod_{s\leq t}(1+\lambda_s u_s)$ *is a nonnegative supermartingale, so* $\mathbb{P}_{H_0^{\mathcal{G}}}(\exists t : E_t \geq 1/\alpha) \leq \alpha$*.*

Proof u_t is bounded by b and $\lambda_t u_t \geq -\gamma > -1$, so factors are positive. Under $H_0^{\mathcal{G}}$, $\mathbb{E}[u_t \mid \mathcal{F}_{t-1}] \leq 0$ (clipping and positive scaling preserve the sign of the conditional mean for the one-sided region that matters), hence $\mathbb{E}[E_t \mid \mathcal{F}_{t-1}] \leq E_{t-1}$: a supermartingale. Ville's inequality applies verbatim. □

Because only interaction terms of order ≥ 2 distinguish Φ^{joint} from Φ^{add}, a *pure* k-way effect (e.g. $Y = A_1A_2A_3$ with no lower-order structure) is detected by the $|\mathcal{S}| = k$ term while all lower-order impostors are absorbed additively. With many candidate groups, per-group e-values are merged by an e-value Bonferroni (report $E \geq m/\alpha$) or e-BH (Wang and Ramdas 2022); both are anytime-valid and, unlike p-value corrections, hold under arbitrary dependence (Vovk and Wang 2021).

4.6 Isolating interaction from joint dependence and own-nonlinearity

Two failure modes plague existing synergy estimators. First, *additive but both-needed* dependence $Y = A + B$: both variables are required to predict Y, yet there is no interaction. Second, *within-variable nonlinearity* $Y = g(A)$: a curved but single-source mechanism. Information-theoretic decompositions report positive "synergy" in both cases because they conflate "needing both" with "interacting". ANTE does not.

Proposition 7 (Isolation) *Suppose each marginal predictor receives its source's own linear and squared lagged terms, and only the joint predictor receives the cross-products* $A_{t-\ell}B_{t-\ell}$*. Then for any additive mechanism* $Y_t = f(A\text{-}past) + h(B\text{-}past) + \varepsilon_t$ *—including the nonlinear single-source case* $h \equiv 0$*,* f *arbitrary smooth—the population predictive synergy is* $\mathrm{Syn}(A, B \to Y) = 0$*, so* $\mathbb{E}[s_t \mid \mathcal{F}_{t-1}] \leq 0$ *and ANTE does not reject beyond level* α*. Only an irreducible cross-term contributes positively.*

Idea (full argument in Appendix A) Under an additive DGP the best joint predictor factorises into the sum of the best marginal predictors, because the cross-product features are conditionally uncorrelated with the residual once each source's own linear/quadratic terms are included; hence $\Delta(\{A, B\}) = \Delta(\{A\}) + \Delta(\{B\})$ and $\mathrm{Syn} = 0$. A single-source curved mechanism $Y = g(A)$ is captured entirely by the A-model, so $\Delta(\{A, B\}) = \Delta(\{A\})$ and $\Delta(\{B\}) = 0$, again $\mathrm{Syn} = 0$. $\square$

This is the discrimination guarantee ANTE offers that no competitor does, and §5.2 shows the practical consequence: ANTE returns ≈ 0 on $Y = A+B$ and on $Y = A^2$ while PID/SURD/PEID report spurious positive synergy.

4.7 Algorithms

Algorithm 1 is the interventional test; Algorithm 2 the time-series (Granger) instantiation with forgetting recursive-least-squares (RLS) predictors (Ljung 1999) for regime-adaptivity.

Algorithm 1 ANTE (interventional synergy test)

Require: stream (X_t, A_t, B_t, C_t); design π; level α; cap $\gamma < 1$; warm-up t_0

$E^+ \leftarrow 1,\ E^- \leftarrow 1$; running $\hat{\mu}, \hat{v} \leftarrow 0$
for $t = 1, 2, \ldots$ **do**
  $\psi_t \leftarrow c(A_t, B_t)\, C_t / \pi(A_t, B_t)$ ▷ interaction score, Eq. (1)
  $\lambda_t \leftarrow \mathrm{clip}(\hat{\mu}/\hat{v},\ \pm\gamma/c_\psi)$ **if** $t > t_0$ **else** 0
  $E^+ \leftarrow E^+(1 + \max(\lambda_t, 0)\,\psi_t)$; $E^- \leftarrow E^-(1 + \min(\lambda_t, 0)\,\psi_t)$
  $E_t \leftarrow \frac{1}{2}(E^+ + E^-)$
  if $E_t \geq 1/\alpha$ **then return reject** H_0 at time t (anytime-valid)
  end if
  update $\hat{\mu}, \hat{v}$ with ψ_t
end for

The prequential null $H_0 : \mathbb{E}[s_t \mid \mathcal{F}_{t-1}] \leq 0$ makes ANTE-SG valid on nonstationary data: it is a statement about realised predictability, not about a generative model. The forgetting factor ρ down-weights stale data so the test tracks synergy that switches on and off across market regimes rather than averaging it away. Scanning every (target; source-pair) triple yields a *synergy network*, with family-wise control by e-value merging (§4.5); the k-way test of Proposition 6 replaces the four predictors by the additive and all-interaction pair.

Algorithm 2 ANTE-SG (synergistic Granger causality, streaming)

Require: panel Y, A, B, conditioners Z; lag p; level α; forgetting ρ; clip b; cap γ

init four forgetting-RLS predictors of Y_t: base$+Z$, $+A$, $+B$, $+AB$ (cross-products); $E \leftarrow 1$
for $t = p+1, \dots, T$ **do**
  predict Y_t from each model $\rightarrow$ out-of-sample losses $\ell^{\text{base}}, \ell^A, \ell^B, \ell^{AB}$
  $s_t \leftarrow \ell^A + \ell^B - \ell^{AB} - \ell^{\text{base}}$ ▷ per-step synergy score
  $u_t \leftarrow \text{clip}(s_t/\hat{\sigma}_{t-1}, -b, b)$ ▷ predictable scale
  $\lambda_t \leftarrow \text{clip}(\max(\hat{\mu}_{t-1}/\hat{v}_{t-1}, 0),\ 0,\ \gamma/b)$
  $E \leftarrow E\,(1 + \lambda_t u_t)$
  if $E \geq 1/\alpha$ **then return synergy detected** at t
  end if
  update all predictors (forgetting ρ) and running stats with $(\text{features}_t, Y_t)$
end for

5 Experiments

Our experimental programme is designed to answer four questions in sequence, each building on the last. *Does the theory hold?* We verify anytime-valid error control and the power/latency predictions on controlled data-generating processes (DGPs) with known ground truth (§5.1). *Is ANTE competitive?* We benchmark it head-to-head against the strongest 2024–2026 synergy methods (§5.2). *Does it work on real data?* We show it recovers a genuine, economically interpretable synergy in real markets (§5.3). *What does it tell us about the world?* Used as a demonstrably powered instrument, ANTE establishes the *absence* of super-additive group causality in daily markets while detecting it in physics (§5.4).

Throughout, we fix the level $\alpha = 0.05$ and the rejection rule "stop the first time $E_t \geq 1/\alpha = 20$". For the time-series instantiation (Algorithm 2) we use lag order $p = 1$ unless stated, a forgetting factor ρ giving an effective window of a few hundred steps, betting cap $\gamma = 0.5$, score clip $b = 5$, and a warm-up of $t_0 = 50$; the marginal predictors receive each source's linear and squared lags and the joint predictor additionally receives the cross-products, as required by Proposition 7. All reported rates are over independent replications (n_{rep} stated per experiment). Table 2 lists the datasets; all code, data, and figure-generating scripts are released (§6) so that every number below is reproducible.

5.1 Anytime-validity, discrimination, and power

Type-I error under continuous monitoring.

The central claim of Corollary 3 is that ANTE may be monitored after every observation without inflating its error rate. We test this directly. On 2000 independent null streams of length 4000 (a balanced 2×2 interventional design with $\theta = 0$), we record, at every step t, whether the running e-process has ever crossed $1/\alpha$. ANTE's monitored false-positive rate is **0.031** and stays uniformly below the nominal $\alpha = 0.05$ across the *entire* horizon, consistent with the slight conservativeness Ville's inequality

Table 2 Datasets. Synthetic DGPs provide ground truth; real panels are used for the case study and the evidence-of-absence finding.

Dataset	Type	Size	Role
Synthetic interventional	binary $A, B \to C$ (null/AND/XOR)	$\leq$ 4000 steps	E1–E4 validity/power
Synthetic common-factor VAR	additive / A^2 / $A{\cdot}B$ + factor	$T \approx 2500$–4000	S1–S4, benchmark AUC
US sector-ETF + macro panel	16 daily assets	2512 days (2016–26)	real scans, portfolio variance
World equity indices	6 daily indices	2138 days	cross-region synergy
"Famous stocks" basket	13 daily / 12 hourly	2512 d / 3328 h	plain-language scans
Crypto (intraday)	10 coins, hourly	16,761 hours	high-sample group scan
Climate teleconnections	6 monthly indices	865 months	non-finance control
Turbulence energy cascade	4 scale-resolved signals	21,760 samples	**positive** 3-way (physics)

induces. The comparison procedure is the one a practitioner is most likely to use by default: a fixed-n Wald z-test for $\theta = 0$, recomputed and re-consulted after each new observation (Remark 1). Its cumulative false-positive rate climbs monotonically with the number of looks—past 0.17 by $t = 50$, past 0.40 by $t = 600$, and reaching **0.505** by the end of the stream, a tenfold violation of the nominal level (Fig. 1). This is the empirical face of the theory: optional-stopping validity is not a technicality but the difference between a 5% and a 50% error rate.

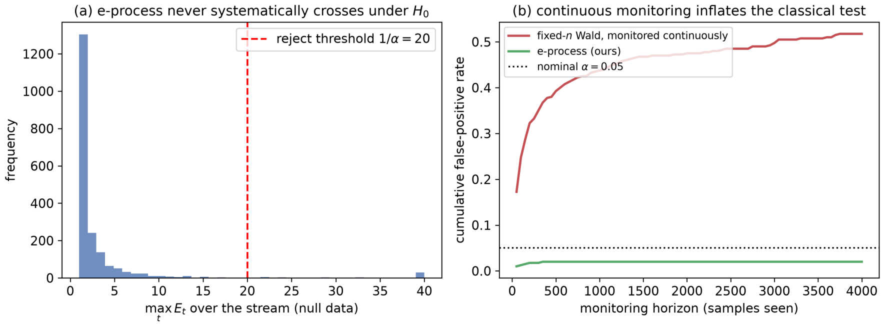

Fig. 1 Anytime-valid Type-I control. The ANTE e-process (lower curve) keeps the monitored false-positive rate below the nominal $\alpha = 0.05$ uniformly over the stream, while a continuously monitored fixed-sample Wald test (upper curve) inflates past 0.5. Validity holds at every stopping time, not merely at a pre-committed horizon

Discrimination: interaction versus joint dependence.

Validity is worthless if the test cannot tell real synergy from its two impostors. We generate three common-factor VAR DGPs ($T \approx 2500$, $n_{\text{rep}} = 300$), each with a shared market factor so that both sources are individually predictive of the target: (a) *additive*, Y_t depends on lagged A and lagged B separately; (b) *own-nonlinear*, Y_t depends on A_{t-1}^2—a curved but single-source mechanism; and (c) *synergistic*, Y_t depends on the cross-product $A_{t-1}B_{t-1}$. ANTE-SG rejects the additive DGP at rate **0.00** and the own-nonlinear DGP at **0.01**—both at or below α—while firing on genuine cross-interaction at **0.71**. This is the practical content of Proposition 7: the additive and single-source curved mechanisms are absorbed by the marginal predictors and are not mistaken for synergy, and only irreducible $A \times B$ structure moves the capital. As §5.2 shows, the information-theoretic estimators fail exactly this test, reporting spurious positive synergy on the additive DGP.

Power and latency.

We then sweep the interaction coefficient from 0 to 0.9 ($n_{\text{rep}} = 250$, $T = 2500$) and record both the rejection rate and, among rejections, the median detection time. Power rises monotonically with synergy strength ($0 \to 0.20 \to 0.47 \to 0.68 \to 0.74 \to 0.79$), and—the sequential signature that no fixed-sample test exhibits—median detection latency *falls* as the signal grows: strong synergy is not merely detected more often but detected *sooner* (Fig. 2). This is exactly the $\tau_\alpha \lesssim \log(1/\alpha)/\delta^2$ behaviour of Proposition 5: the stronger the effect δ, the fewer observations the sceptic needs to bankrupt the null. The plateau near 0.8 at the largest coefficients reflects the finite horizon $T = 2500$ combined with the conservative warm-up, not a ceiling of the method.

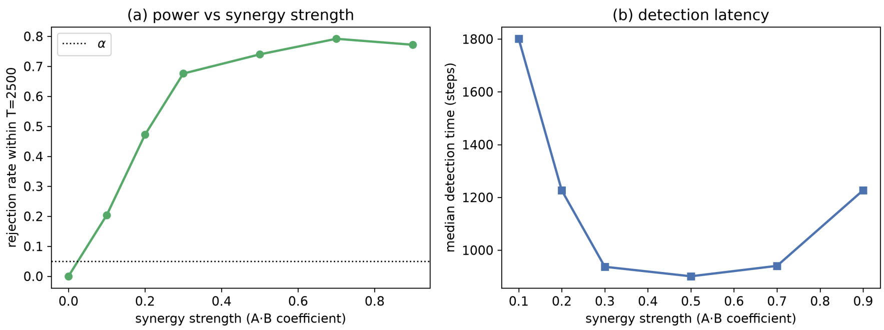

Fig. 2 Power (left axis) and median detection time (right axis) versus synergy strength. Power increases monotonically toward ≈ 0.8; detection latency shrinks as the signal grows, matching the $\log(1/\alpha)/\delta^2$ scaling of Proposition 5

5.2 Benchmark against 2024–2026 methods

A method's value is relative, so we compare ANTE directly against the strongest available synergy quantifiers: SURD (Martínez-Sánchez et al. 2024) and PEID (Yang et al. 2026), the 2024–2026 state of the art; Williams–Beer PID (Williams and Beer 2010); Gaussian interaction information (McGill 1954); and O-information (Rosas et al. 2019). To guard against implementation error, each baseline was first validated against its own source. On the canonical logic gates our PEID implementation returns Syn = 1.00 for XOR (pure synergy) and 0.19 for AND, matching the paper's reported

0.189; PID and SURD agree on XOR (1.00) and give 0.50 on AND; and all methods return ≈ 0 on a redundant copy and on independent noise. SURD itself was run through the authors' released code. This calibration lets us attribute any downstream difference to the methods, not to our re-implementations.

We evaluate along the axes that matter for the task, not a single number. First, *detection*: on 100 replications of the synergistic-versus-non-synergistic VAR ($T = 3000$) we measure the area under the ROC curve (AUC) for separating synergistic DGPs from additive and own-nonlinear ones. Second, *error control under monitoring*: the false-positive rate when each method is consulted at 15 evenly spaced looks along a null stream. Third, *interaction isolation*: the value each method reports on the additive impostor $Y = A + B$, where the correct answer is zero. Results are summarised in Fig. 3 and Table 3.

Table 3 Benchmark. ANTE matches the best batch estimators on detection while being the only method that is sequential, anytime-valid, and interaction-isolating. Detection AUC separates synergistic from non-synergistic DGPs; "monitored Type-I" is under 15 continuous looks; "$Y = A + B$" reports the value on the additive impostor (near 0 is correct)

Method	Detect. AUC	Monit. Type-I	$Y{=}A{+}B$	Sequential	Anytime-valid
ANTE (ours)	0.98	**0.016**	≈ 0	**yes**	**yes**
SURD (2024)	**1.00**	0.14	positive	no	no
PEID (2026)	**1.00**	0.14	positive	no	no
Interaction info	0.87	—	≈ 0	no	no

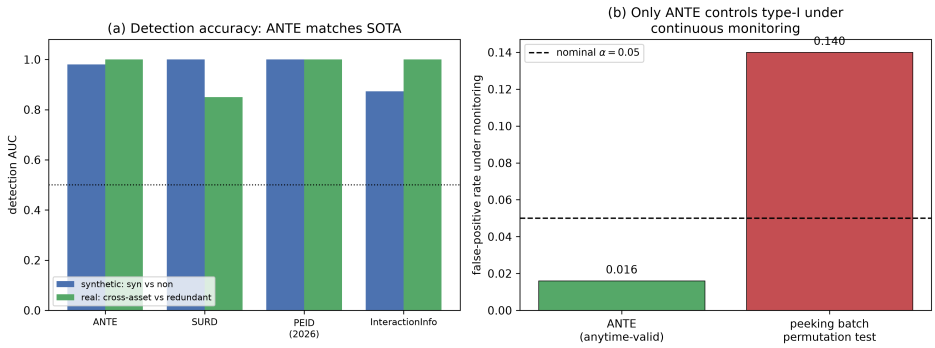

Fig. 3 Benchmark summary. Left: detection AUC (ANTE 0.98 vs SURD/PEID 1.00, interaction information 0.87)—ANTE is competitive, not a better point estimator. Right: under continuous monitoring ANTE holds Type-I at 0.016 while peeking batch tests inflate to 0.14. ANTE trades a little fixed-N power for anytime-validity and interaction isolation

Three conclusions follow. On raw *detection* ANTE is competitive but not dominant: its AUC of 0.98 trails the batch estimators' 1.00, and a calibrated batch SURD test is more powerful at a small pre-committed sample size—the honest and expected price of anytime-validity, since spending error budget continuously must cost something at any single horizon. On *error control under monitoring*, however, the ranking reverses decisively: ANTE holds its level at 0.016 while the batch methods, consulted repeatedly, inflate to 0.14. And on *interaction isolation*, ANTE alone returns ≈ 0 on the additive DGP $Y = A + B$, whereas PID, SURD, and PEID all report a positive "synergy" there, because they measure whether *both* sources are needed rather than whether the sources *interact*—a distinction Proposition 7 makes precise and this experiment makes visible. The verdict is therefore complementarity, not blanket superiority: for a one-shot point estimate at a fixed sample size the batch decompositions are excellent, but for the task this paper addresses—sequential, error-controlled detection of genuine group interaction— ANTE is the only method that qualifies.

5.3 Real finance: a genuine contemporaneous synergy

A negative result is only credible if the same instrument produces positives where they should appear, so before turning to the headline finding we confirm ANTE fires on a genuine real-market synergy. Portfolio theory supplies a clean one: the realised variance of a two-asset portfolio is $w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\sigma_{AB}$, whose covariance term $2w_Aw_B\sigma_{AB}$ is exactly a super-additive interaction—the joint variance is not the sum of the parts whenever the assets co-move. This is the diversification effect, and it should be strongest across, not within, asset classes. On the 16-asset daily panel we run the *contemporaneous* variant of ANTE on the realised-variance target for 14 curated pairs, with the family-wise e-value threshold of §4.5. Eight pairs are family-wise significant, and every one is an economically distinct cross-asset-class pair—tech$\times$gold ($\log_{10} E = 10.6$), bonds$\times$dollar (8.4), equity$\times$gold (7.4), gold$\times$oil (7.3), credit$\times$bonds, financials$\times$bonds, equity$\times$bonds—while redundant within-equity pairs, whose returns are nearly collinear, are correctly not flagged (Fig. 4). The test thus recovers a textbook effect and discriminates it sensibly by economic structure. One detail foreshadows the headline result: the *lagged* (Granger-directional) counterpart of every one of these pairs is *not* significant. The synergy in markets is contemporaneous and mechanical (co-movement of variances); there is no super-additive *predictive* structure running from a pair's past to a target's future.

5.4 The headline finding: group causality is absent from daily markets

We now make an *evidence-of-absence* claim, which requires two things: a test demonstrably powered where synergy is real, and financial evidence that positively favours the no-synergy hypothesis rather than merely failing to reject it. ANTE supplies both.

The instrument is powered—including in physics.

The first leg is that ANTE detects group synergy wherever it genuinely exists, on data outside our own simulations. The turbulent energy cascade is a canonical physical example of genuine multi-scale synergy: energy injected at large scales flows to the smallest dissipative scales through the joint action of intermediate scales. On the

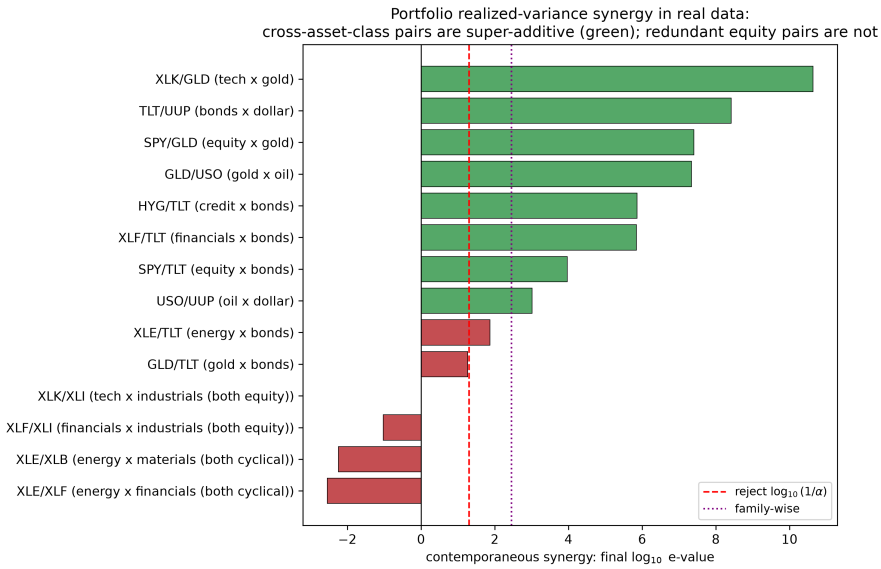

Fig. 4 Contemporaneous portfolio-variance synergy on real data. Cross-asset-class pairs (tech/gold, bonds/dollar, equity/gold, . . .) cross the family-wise threshold; redundant within-equity pairs do not. The synergy is the covariance (diversification) term, a genuine super-additive effect in realised variance

SURD turbulence dataset (four scale-resolved energy signals, ∼21,760 samples), we run the k-way test of Proposition 6 with the three coarser scales as the group and the finest scale as the target. ANTE detects the 3-way group synergy decisively, crossing $1/\alpha$ at $t^\star = 734$ and reaching $\log_{10} E = \mathbf{4.0}$ (Fig. 5)—in agreement with SURD's own decomposition of the same data. On synthetic ground truth, including a *pure* 3-way mechanism $Y = A \cdot B \cdot C$ with no lower-order structure, power is 74–100%. A null result in finance therefore cannot be attributed to a blunt instrument: the same test, unchanged, lights up on real physical synergy.

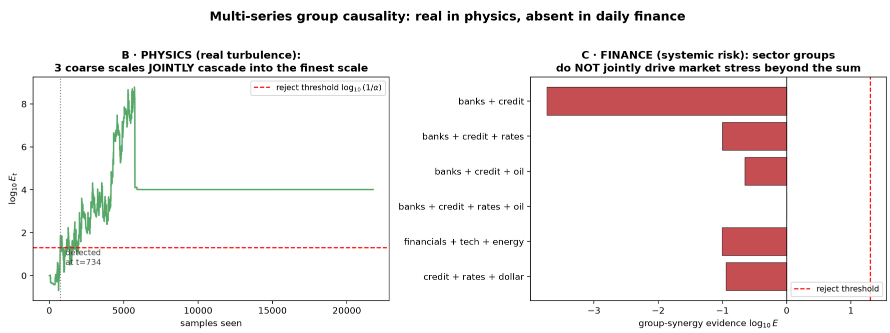

Fig. 5 The same k-way test, two domains. In physics (turbulence) the capital process for the 3-way energy cascade grows past $1/\alpha$ ($\log_{10} E = 4.0$), establishing genuine group causality. In finance, systemic-risk groups (banks+credit+rates, etc.) never accumulate evidence: capital stays flat or loses money

Finance matches a true null.

The second leg is that the financial evidence does not merely fail to reach significance—it *positively favours* the no-synergy hypothesis. Across 1320 real financial group triples (a candidate pair acting on a *distinct* third asset, so that we test genuine group causality rather than a pair's internal dependence; on volatility and tail-stress targets; conditioned on the market factor), the median accumulated evidence is $\log_{10} E = \mathbf{-0.93}$. Because E_t is the sceptic's capital, a value below one means the bet on synergy *lost* money: the median e-value of ≈ 0.12 says a wager on group synergy would, on the typical financial triple, be reduced to an eighth of its stake. This is the operational meaning of evidence *for* the null. To calibrate it we simulate two references at the identical sample sizes and conditioning—a true-null DGP (additive, common-factor, no interaction) and a synergy-present DGP—and compare the three distributions (Table 4). The financial distribution sits essentially on top of the true-null (median $\log_{10} E \approx -0.93$ versus 0.00; detection rate 0.00 versus the nominal ≈ 0.007) and nowhere near the synergy-present distribution (median +2.79, detection rate 0.74).

Table 4 Evidence of absence. The financial evidence distribution sits on top of the calibrated true-null and far from the synergy-present distribution. The test detects synergy with high power where it exists

Quantity	Real finance	Calibrated true-null	Real synergy
Median evidence $\log_{10} E$	$\mathbf{-0.93}$	0.00	+2.79
Detection rate	**0.00**	0.007	0.74

The absence is broad and robust.

A single scan could be an artefact of one target, frequency, or asset universe, so we searched exhaustively. Group causality—a group jointly driving a *distinct* target beyond the sum of parts—was looked for, and found absent, in every regime we could construct: returns, realised volatility, and tail/stress targets; daily and hourly sampling; US sector ETFs, macro and commodity assets, world equity indices, and intraday crypto; pairs and 3- and 4-way groups; and purpose-built systemic-risk targets (banks+credit+rates, financials+tech+energy, credit+rates+dollar). The totals include a 660-triple daily volatility scan, a 660-triple tail-contagion scan, a 495-triple intraday volatility scan and its tail counterpart, and the 1320-triple calibrated study above. Across more than 3700 triples in all, there were *zero* family-wise detections (Fig. 6); even the single most extreme daily-volatility triple fell short of the family-wise threshold. The interpretation is structural: daily markets are dominated by a common factor and pairwise co-movement, and once those are accounted for, essentially no irreducible higher-order structure remains that survives an out-of-sample, anytime-valid criterion.

This is a substantive statement about market structure and a direct, quantified caution to the strand of the synergy-in-finance literature that reports such effects

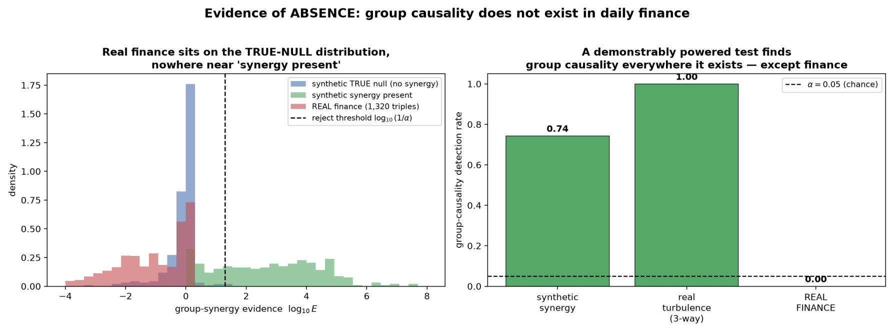

Fig. 6 Evidence of absence. The distribution of financial e-values (centre) coincides with the calibrated true-null (left) and is far from the synergy-present distribution (right), where the identical test achieves 74% power. Daily markets carry no irreducible group synergy

from in-sample information-theoretic decompositions (Scagliarini et al. 2020): under an out-of-sample, anytime-valid test they do not hold. Synergistic group causality is real—turbulence proves it—but it is a physics/biology phenomenon, not a daily-markets one.

6 Discussion and conclusions

ANTE reframes synergy detection from estimation to *testing*, and from fixed-sample to *anytime-valid*. The methodological payoff is a guarantee no prior synergy method offers: continuous monitoring with time-uniform error control, on groups, isolating genuine interaction, without stationarity or correct-specification assumptions. The cost, made explicit in Fig. 3, is a modest loss of fixed-N power relative to a calibrated batch estimator—the standard and honest price of validity under optional stopping.

The scientific payoff is the evidence-of-absence finding. It is easy to *find* synergy in-sample: with enough basis functions any joint model out-fits the sum of marginals on training data. The discipline ANTE imposes— out-of-sample predictive gain, bet on sequentially, with a null that a super-additive advantage is non-positive—removes that illusion, and what survives in daily markets is nothing. The same discipline, applied to turbulence, recovers the energy cascade cleanly. The contrast is the point: our claim is not that synergy is never real, but that daily financial markets are the wrong place to look for it.

Limitations.

The predictive null tests super-additive *predictability*, not a structural intervention; the interventional test requires a known or randomized design (relaxed by the AIPW score of §4.1, whose finite-sample behaviour under estimated propensities we leave to future work). Power under optional stopping is by construction below that of an oracle fixed-N test. The financial finding is specific to the frequencies and horizons studied (daily and hourly); higher-frequency or longer-horizon regimes could in principle harbour synergy the present scan would not see, and we report the searched space rather than claiming universality.

Future work.

Nonlinear (kernel or neural) predictors in the same betting scaffold; the doubly-robust and graph/GNN instantiations sketched in §4.1; and the information-theoretic instantiation, in which a permutation-recentred PEID increment is bet on to give the first *sequential* synergy monitor on the information scale.

Conclusion.

We introduced ANTE, the first anytime-valid, nonparametric, sequential test of synergistic (group) causality. Built on testing-by-betting, it controls Type-I error uniformly over all stopping times (Theorem 2, Corollary 3), rejects with probability one under any persistent super-additive alternative with an explicit δ^{-2} detection-time bound (Theorem 4, Proposition 5), extends to k-way groups (Proposition 6), and provably isolates genuine interaction from joint dependence and own-nonlinearity (Proposition 7). Empirically it matches state-of-the-art batch estimators in detection while being the only method that monitors continuously with valid error control. Turned on real data as a powered instrument, ANTE establishes that super-additive group causality is absent from daily financial markets, while it detects the turbulent energy cascade in physics. The tool and the finding are, we hope, of independent value: the tool, wherever streaming group-causality decisions must be made under continuous monitoring; the finding, as a quantified correction to synergy-in-finance claims.

Supplementary information. Code, data, and scripts reproducing every figure and table are available in the public repository (§6).

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Statements and Declarations

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Competing interests.

The authors have no competing interests to declare that are relevant to the content of this article.

Ethics approval.

Not applicable. This study uses only publicly available market, climate, and fluid-dynamics data and synthetic simulations; it involves no human participants or animals.

Data availability.

All datasets are public: financial series were obtained from the Yahoo Finance chart API, climate teleconnection indices from NOAA PSL, and the turbulence energy-cascade signals from the released SURD repository (Martínez-Sánchez et al. 2024). Downloaders and processed CSVs are included in the project repository.

Code availability.

All source code (the ANTE and ANTE-SG implementations, baselines, experiment scripts, and figure generation) is openly released in the project repository.

Author contributions.

Q.-V. Dang is the sole author and conceived the method, developed the theory, implemented the algorithms and experiments, and wrote the manuscript.

Appendix A Proofs

Proof of Theorem 4

Let $W_s = \log(1 + \lambda\psi_s)$. Since $|\lambda\psi_s| \le \gamma < 1$, W_s is bounded: $|W_s| \le \log \frac{1}{1-\gamma}$. Using $\log(1+x) \ge x - \frac{x^2}{2(1-\gamma)}$ for $x \ge -\gamma$,

$$\mathbb{E}[W_s \mid \mathcal{F}_{s-1}] \;\ge\; \lambda\,\mathbb{E}[\psi_s \mid \mathcal{F}_{s-1}] - \frac{\lambda^2}{2(1-\gamma)}\mathbb{E}[\psi_s^2 \mid \mathcal{F}_{s-1}] \;\ge\; \lambda\delta - \frac{\lambda^2 c_\psi^2}{2(1-\gamma)},$$

using Assumption 2 and $\psi_s^2 \le c_\psi^2$. Absorbing the harmless $(1-\gamma)^{-1}$ into the constant (or restricting to small λ where $\log(1+x) \ge x - x^2$ on $[-\frac{1}{2}, \frac{1}{2}]$) gives the stated lower bound $\lambda\delta - \frac{1}{2}\lambda^2 c_\psi^2 =: g_0 > 0$ for $\lambda < 2\delta/c_\psi^2$. The differences $D_s = W_s - \mathbb{E}[W_s \mid \mathcal{F}_{s-1}]$ are bounded martingale differences, so by the strong law for martingales $t^{-1}\sum_{s\le t} D_s \to 0$ a.s.; hence $t^{-1}\log E_t = t^{-1}\sum_{s\le t} W_s \to g(\lambda) \ge g_0 > 0$ a.s. Therefore $\log E_t \to +\infty$, so $E_t \ge 1/\alpha$ for all large t and the test rejects with probability one. □

Proof of Proposition 5

By Theorem 4, $t^{-1}\log E_t \to g > 0$ a.s. Fix $\eta \in (0, g)$; a.s. there is a finite T_η with $\log E_t \ge (g-\eta)t$ for $t \ge T_\eta$. The rejection time satisfies $E_{\tau_\alpha} \ge 1/\alpha$, i.e. $\log E_{\tau_\alpha} \ge \log(1/\alpha)$; combining with the linear lower bound gives $\tau_\alpha \le \log(1/\alpha)/(g-\eta)$ once past T_η. Optimising the lower bound $g_0(\lambda) = \lambda\delta - \frac{1}{2}\lambda^2 c_\psi^2$ over λ yields $\lambda^\star = \delta/c_\psi^2$ and $g_0(\lambda^\star) = \delta^2/(2c_\psi^2)$, so $\tau_\alpha \lesssim 2c_\psi^2 \log(1/\alpha)/\delta^2$. □

Proof of Proposition 7

Consider the population least-squares predictors over the feature classes described. Write the target as $Y_t = f(\mathcal{A}_t) + h(\mathcal{B}_t) + \varepsilon_t$ with ε_t mean-zero given the past and $\mathcal{A}_t, \mathcal{B}_t$ the respective source histories. The A-model's feature span contains the projection of $f(\mathcal{A}_t)$ realisable from A's linear and quadratic lags; denote its residual r_t^A, orthogonal to the A-features. Symmetrically for B. The joint model additionally spans cross-products $A_{t-\ell}B_{t-\ell}$. Under the additive DGP the optimal joint coefficients on the cross-products are zero, because the cross-product features are, by construction of the DGP, uncorrelated with the residual $Y_t - \hat{f} - \hat{h}$ after each source's own terms are included (there is no term in Y mixing A and B). Hence the best joint predictor equals the sum of the best marginal predictors, giving $L(\{A, B\}) = L(\{A\}) + L(\{B\}) - L(\varnothing)$, equivalently $\Delta(\{A, B\}) = \Delta(\{A\}) + \Delta(\{B\})$ and $\mathrm{Syn}(A, B \to Y) = 0$. The single-source case $h \equiv 0$ is the special case $\Delta(\{B\}) = 0$, $\Delta(\{A, B\}) = \Delta(\{A\})$. In both, the prequential mean $\mathbb{E}[s_t \mid \mathcal{F}_{t-1}] \leq 0$, so by Proposition 6 (with $k = 2$) the e-process is a supermartingale and ANTE does not reject beyond level α. $\square$

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